

Certification in Agentic Systems and Design (IITRAS-260113)

Final Offline Evaluation Syllabus

Module 1: Agentic Foundation & Architecture

S. No.	Lecture Title	Topics
1	Defining Agentic Systems	Agent definition ♦ autonomy vs automation ♦ perception-reasoning-action loop ♦ agents vs chatbots vs copilots
2	Core Components of AI Agents	Planning ♦ reasoning ♦ memory ♦ tool use ♦ environment interaction ♦ feedback loops
3	Exploring Agent Behavior	Interacting with Claude/GPT agents ♦ observing reasoning traces ♦ identifying components in action
4	Business Case and Use Cases for Agents	Business value ♦ ROI analysis ♦ customer support agents ♦ financial analysis ♦ HR automation ♦ content creation ♦ research assistants
5	Understanding the LLM Layer: How LLMs Work Internally	Neural network foundations ♦ pretraining objectives ♦ next-token prediction ♦ training data and scale ♦ emergent capabilities
6	Transformer Fundamentals	Attention mechanism ♦ self-attention ♦ multi-head attention
7	LLM Selection for Agents	Model comparison (Claude, GPT-4, Gemini, Llama, Mistral) ♦ capability trade-offs ♦ cost vs latency vs quality
8	Prompt Engineering Fundamentals	System prompts ♦ role definition ♦ constraint specification ♦ output formatting ♦ few-shot prompting
9	Prompt Engineering Hands On	Writing agent personas ♦ capability definition ♦ testing variations ♦ measuring effectiveness

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10	Python Essentials-1	Python data structures ♦ functions and classes
11	Python Essentials-2	Classes & Objects ♦ OOPS ♦ Exception Handling ♦ Numpy
12	Pydantic for Data Validation	Pydantic models ♦ field validators ♦ nested models ♦ serialization ♦ settings management
13	Python and Pydantic Hands-on	Building data models ♦ validation exercises ♦ async patterns ♦ type-safe components
14	FastAPI Fundamentals	FastAPI basics ♦ path operations ♦ request/response models ♦ dependency injection ♦ automatic docs
15	FastAPI & Databases	Database integration with FastAPI ♦ CRUD operations
16	FastAPI Advanced Patterns	Async endpoints ♦ middleware ♦ background tasks ♦ error handling
17	WebSockets in FastAPI	WebSocket support in FastAPI ♦ streaming
18	Introduction to Agent Frameworks	LangGraph ♦ LangChain ♦ CrewAI ♦ AutoGen ♦ n8n ♦ Google ADK ♦ Vercel AI SDK ♦ OpenAI Agents SDK ♦ framework comparison

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Module 2: Agent Components - Memory, Tools & RAG

S. No.	Lecture Title	Topics
19	Advanced Prompt Engineering for Agents	chain-of-thought prompting ♦ structured prompts ♦ reasoning prompts
20	Self-Reflection and Feedback Loops	self-correction prompts ♦ iterative prompting ♦ agent prompt design
21	Introduction to Memory in AI Agents	why agents need memory ♦ stateless vs stateful agents ♦ types of agent memory
22	Short-Term vs Long-Term Memory	conversational memory ♦ persistent memory ♦ memory storage strategies ♦ (Main focus on Therotical discussion)
23	Introduction to Databases for AI Systems	relational databases ♦ SQL basics ♦ structured vs unstructured data
24	Using SQL Databases with AI Applications	querying databases ♦ database access patterns for agents
25	Embeddings and Semantic Representation	text embeddings ♦ semantic similarity
26	Introduction to Vector Databases	vector indexing ♦ similarity search
27	Implementing Vector Search Systems	storing embeddings ♦ querying vector DB
28	Introduction to RAG	limitations of LLM knowledge ♦ grounding responses
29	RAG Architecture and Pipeline	retriever ♦ generator ♦ knowledge sources
30	Building a RAG Pipeline	document loaders ♦ chunking
31	Evaluating and Improving RAG Systems	hallucination reduction ♦ retrieval tuning
32	Working with APIs and JSON	REST APIs ♦ JSON structures ♦ API responses
33	Tool Integration in AI Agents	function calling ♦ connecting APIs and tools

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Module 3: Hands-On Single-Agent Development & Use Cases

S. No.	Lecture Title	Topics
34	Ollama: Exploring Another World of LLMs	Ollama install ♦ light model ♦ Python API ♦ Ollama Cloud ♦ dual-mode script
35	Introduction to LangChain: Concepts Architecture and First Demo	definition ♦ why framework ♦ stack placement ♦ Runnables ♦ modules overview ♦ Core vs Community ♦ PromptTemplate ♦ LCEL ♦ output parsers ♦ instructor demo chain
36	LangChain Environment Setup and First LCEL Chain	venv ♦ packages ♦ .env ♦ folders ♦ ChatOllama ♦ ChatPromptTemplate ♦ LCEL ♦ StrOutputParser ♦ hello_chain.py
37	LangChain Tools: Custom Tools and Tool Calling	@tool ♦ bind_tools ♦ tool_calls ♦ ToolMessage ♦ error handling
38	Building Your First LangChain Agent	create_tool_calling_agent ♦ AgentExecutor ♦ max_iterations ♦ test pack
39	LangChain Memory on Agents	MessagesPlaceholder ♦ chat_history ♦ multi-turn script ♦ message append
40	LangChain RAG Pipeline	loaders ♦ Chroma ♦ retriever ♦ LCEL RAG chain ♦ grounding comparison
41	RAG Tool and Integrated LangChain Agent	create_retriever_tool ♦ second tool ♦ multi-turn ♦ eval pack
42	Evaluating LangChain Agents: Test Sets and Logging	eval JSON ♦ runner ♦ results.csv ♦ failure trace
43	Debugging and Iterating LangChain Agents	failure class ♦ prompt/tool patch ♦ retrieval tune ♦ quality metrics
44	Hands-On Real-World Use Cases	finance use-case pattern ♦ HR onboarding implementation ♦ content use-case pattern ♦ integrated agent extension ♦ eval harness ♦ live demo ♦ module checklist

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Module 4: Multi-Agent Collaboration and Deployment Strategy

S. No.	Lecture Title	Topics
45	Multi-Agent Architecture HTTP and Automation Foundations	single-agent vs multi-agent ♦ task decomposition ♦ role-based agents ♦ sequential vs collaborative workflows ♦ researcher-writer-editor pipeline ♦ HTTP methods ♦ triggers ♦ webhooks ♦ API automation
46	Introduction to n8n Workflow Automation	n8n workspace ♦ triggers ♦ nodes ♦ connections ♦ expressions ♦ credentials ♦ first workflow
47	n8n LLM Integration and AI Workflow Nodes	LLM nodes ♦ prompt configuration ♦ HTTP Request node ♦ chaining AI steps ♦ error branches
48	Building End-to-End AI Automation Pipelines with n8n	document ingestion ♦ summarization ♦ routing ♦ notifications ♦ pipeline testing ♦ workflow export
49	CrewAI: Roles Tasks and First Multi-Agent Crew	Agent ♦ Task ♦ Crew ♦ Process ♦ role-task-crew model ♦ tools per agent ♦ crew kickoff ♦ output artifacts
50	CrewAI: End-to-End Multi-Agent Workflow	custom tools ♦ hierarchical process ♦ sequential process ♦ memory optional ♦ output validation ♦ iteration
51	AutoGen: Conversable Agents and Tool Use	AssistantAgent ♦ UserProxyAgent ♦ conversable-agent model ♦ register_function ♦ code execution optional ♦ termination conditions
52	AutoGen: Group Chat and Multi-Agent Orchestration	GroupChat ♦ GroupChatManager ♦ speaker selection ♦ max rounds ♦ human input optional ♦ multi-agent handoffs
53	make.com: No-Code AI Automation Scenarios	scenarios ♦ modules ♦ routers ♦ OpenAI or HTTP modules ♦ data stores ♦ scheduling ♦ error handling
54	ChatGPT Agent and Hosted Agent Builder Patterns	ChatGPT Agent ♦ knowledge sources ♦ actions ♦ instructions ♦ guardrails ♦ hosted vs self-hosted trade-offs
55	LLM Operations, Security and Guardrails for Agent Systems	prompt and config versioning ♦ production eval gates ♦ token and cost tracking ♦ secrets and access ♦ PII handling ♦ input output guardrails ♦ policy layers ♦ human-in-the-loop
56	Deployment and Monitoring for Agent Systems	strategising deployment ♦ hosting and runtime choices ♦ environments ♦ observability ♦ monitoring workflows ♦ logging agent decisions ♦ trace and audit fields ♦ performance tracking ♦ incident response planning
57	Governance, Ethical Scaling and Cost Control for Agent Systems	AI governance ♦ data privacy ♦ bias and safety ♦ human oversight ♦ cost controls ♦ policies ♦ audit trails
58	Designing a Multi-Agent System for Business	workflow diagram ♦ agent roles ♦ handoffs ♦ tool and data map ♦ risks ♦ success metrics ♦ finance HR or content scenario

Note:

All the topics covered in the live sessions will be considered as part of the evaluation syllabus.